Structural Relation Enhancement for Open-Vocabulary Pedestrian Attribute Recognition

Jiandong Jin, Chenglong Li*, *Senior Member, IEEE*, Xiao Wang*, Jin Tang

***Abstract*—Open-Vocabulary Pedestrian Attribute Recognition (OV-PAR) aims to recognize and understand novel attributes that do not appear in the training set. Although much progress has been made in existing open-vocabulary recognition methods for natural scenes, there are two major challenges in fine-grained attribute recognition of pedestrian images. (1) Intra-group pedestrian attributes often lie in local regions with minimal semantic differences (e.g., sports and casual shoes) and are thus difficult to distinguish. (2) Inter-group attributes exhibit significantly different semantic hierarchies (e.g., action vs. shoes), requiring the model to balance global understanding and fine-grained discernment. To address the above challenges, we propose a Structural Relation Enhancement Network (SRENet) that jointly models spatial, semantic, and attribute structural relations to capture local discriminative details, global contextual information, and common knowledge among intra-group attributes, thereby enhancing generalization ability for OV-PAR. Specifically, we perform attribute-fair spatial-semantic alignment and multi-level semantic fusion to distinguish subtle attribute differences while maintaining global context understanding, thereby preventing overfitting to particular attribute types. Moreover, we propose an intra-group knowledge sharing sentence retrieval paradigm that allows novel attributes to inherit base attribute representations, thereby improving generalization to novel attributes. This work presents a systematic study of OV-PAR and introduces two benchmarks, evaluating twenty representative methods. Extensive experiments demonstrate that our SRENet significantly outperforms state-of-the-art methods. The code and benchmark will be released on https://github.com/Event-AHU/OpenPAR.**

***Index Terms*—Pedestrian Attribute Recognition, Multi-Modal Learning, Vision-Language, Open-Vocabulary Recognition**

I. Introduction

PEDESTRIAN attribute recognition (PAR) [1] aims to infer semantic attributes of pedestrians from images or videos and serves as a key component in person detection [2], tracking [3], retrieval [4], [5], and re-identification [6], [7]. Recent methods [8]–[11] have achieved strong performance [12]–[14] by leveraging advanced backbones and multimodal cues [15], [16]. However, as shown in Fig. 1 (a), their fixed attribute vocabularies and closed-set classifiers [17]–[19] limit generalization to real-world scenarios with novel or evolving attributes.

With the rise of vision-language models, it becomes promising to replace closed-set attribute classifiers with language-conditioned recognition, enabling the model to infer unseen attributes described by text. Nevertheless, OV-PAR is not a plug-and-play extension of generic open-vocabulary recognition due to a fundamental task mismatch: prior open-vocabulary recognition in natural scenes [20], [21] mainly targets instance-level understanding of multiple objects, whereas OV-PAR demands fine-grained reasoning over a single pedestrian. As shown in Fig. 1, it further requires part-aware evidence grounding (e.g., shirt vs. suit vs. striped top) and must handle attributes spanning diverse semantic levels and spatial scales. For example, body-shape attributes rely on global cues, while shoe attributes depend on small, localized regions.

Based on these observations, we summarize four practical obstacles for OV-PAR: 1) The model should be able to align novel attributes with the corresponding visual evidence, enabling recognition of unseen attributes that closed-set classifiers cannot readily achieve. 2) Attributes differ substantially in semantic granularity (e.g., poses rely on global cues, whereas shoes depend on fine local details). This requires the model to balance global context and local discriminative cues, avoiding bias toward particular attribute types. 3) Fine-grained attributes often exhibit only slight differences that hinge on localized cues (e.g., shirts vs. vests mainly differ in the presence of sleeves), and without strong spatial localization and local cue mining, these signals can be missed, leading to incorrect distinctions. 4) Existing OV-PAR benchmarks do not yet support standardized comparisons with mainstream open-vocabulary recognition methods.

To address these challenges, we propose a Structural Relation Enhancement Network (SRENet) for OV-PAR. Its design follows two related properties of pedestrian attributes: attributes within a group share visual regions but differ in subtle cues, while attributes across groups require evidence at different semantic levels. These properties motivate joint modeling of localized discrimination, attribute-dependent semantic aggregation, and group-level knowledge transfer. In particular, the Spatial Structure Enhancement (SpaSE) branch leverages the visual-language alignment priors embedded in CLIP. It employs a class-balanced optimal transport mechanism to allocate fair and precise attention regions for each attribute. This design captures fine-grained discriminative cues while ensuring balanced visual guidance for both base and novel classes. The Semantic Structure Enhancement (SemSE) branch adaptively fuses high-level contextual semantics with low-level structural details. This adaptive fusion balances fine-grained perception and global context understanding across attributes of different semantic levels. Finally, to exploit the regional similarity among intra-group attributes, we design an

Jiandong Jin, Xiao Wang and Jin Tang are with the School of Computer Science and Technology, Anhui University, Hefei 230601, China. (email: jdjinahu@foxmail.com, xiaowang@ahu.edu.cn, tangjin@ahu.edu.cn)

Chenglong Li is with Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui Provincial Key Laboratory of Multimodal Cognitive Computation, the School of Artificial Intelligence, Anhui University, Hefei 230601, China. (email: lcl1314@foxmail.com)

* Corresponding Author: Chenglong Li, Xiao Wang

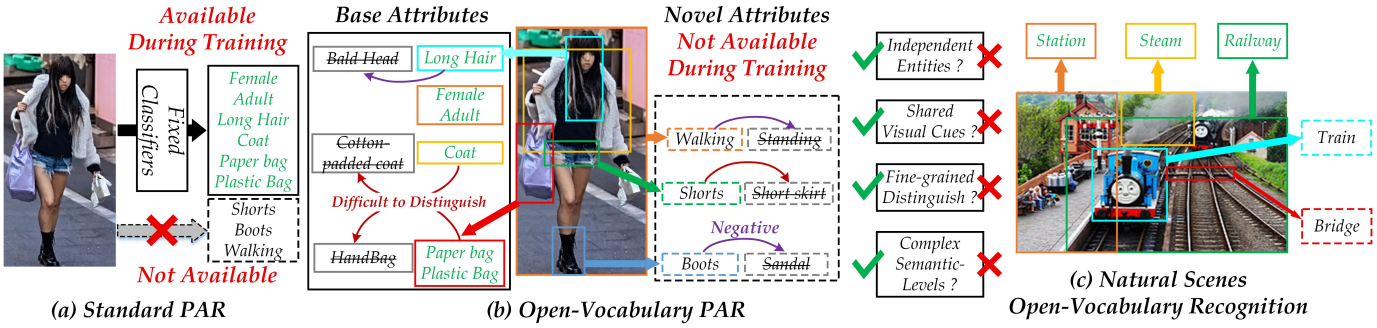


Fig. 1. This figure illustrates the core distinctions between OV-PAR, standard PAR, and open-vocabulary recognition in natural scenes. (a) In standard PAR, all attributes predicted during testing must have precise annotations available during training. (b) OV-PAR eliminates the dependence on predefined independent classifiers and instead explores the association between visual features and attribute semantics, enabling generalization to unseen attributes. (c) Compared with open-vocabulary multi-label recognition in natural scenes, the pedestrian scenario focuses on fine-grained reasoning for a single entity, where intra-group attributes correspond to highly similar visual regions (e.g., *paper bag* vs. *handbag*). In contrast, inter-group attributes exhibit complex semantic hierarchies (e.g., *walking* involves the entire body, whereas *paper bag* corresponds only to the hand region).

Attribute Structure Enhancement (AttSE) branch. Specifically, all attributes are divided into several semantic groups (e.g., head, upper-body clothing, lower-body clothing) to construct a group-level attribute structure. This grouping facilitates the sharing of visual cues within each group, allowing novel classes to inherit the learned visual knowledge from base classes and strengthening intra-group dependency modeling.

SRENet integrates complementary evidence from spatial locations, encoder layers, and intra-group label configurations within a shared CLIP semantic space for unified inference over an open attribute vocabulary. During training, the attribute-level losses of SpaSE and SemSE jointly optimize their shared visual and attribute text prompts, enabling the learned representations to distinguish visually similar attributes through subtle local differences and recognize abstract attributes, such as age and action, using multi-level contextual information. SemSE and AttSE further share a multi-level visual aggregation module: SemSE uses attribute text queries to aggregate visual information relevant to each attribute across layers, while AttSE uses group prototypes to aggregate visual evidence shared by attributes within each group, producing group-specific representations. Group-level contrastive learning (GCL) aligns these group representations with sentence representations generated from base labels, supervising the shared aggregation module and sentence text prompts so that group-level supervision also shapes attribute-level visual aggregation. At inference, AttSE uses the group representations produced by this shared module to retrieve sentences describing configurations that include novel attributes, transferring visual knowledge learned under base-attribute supervision to novel-attribute recognition. Scores from the structural branches and the global branch are mapped to a common attribute order and averaged to obtain the final prediction.

POAR [22] constructs cross-domain OV-PAR benchmarks by integrating multiple PAR datasets. However, the merged benchmark combines dataset-specific annotation protocols, derives attribute groups mainly from label intersections and differences, and covers a limited set of zero-shot and open-vocabulary methods. We therefore construct two unified OV-PAR benchmarks from MSP60K [23] and RAPv2 [24], using consistent annotation rules and explicit class partitions. They respectively emphasize evaluation across diverse environments and richer attribute vocabularies. We apply standard open-vocabulary metrics and evaluate 20 representative methods to provide a standardized platform for OV-PAR research.

The contributions of this paper can be summarized as follows:

- We propose a novel OV-PAR framework which jointly models spatial, semantic, and attribute structural relations to capture both fine-grained details and global context, while transferring learned knowledge to novel classes.
- We jointly design the SpaSE and SemSE branches to balance fine-grained discrimination among similar attributes and mitigate discrepancies across semantic levels.
- We propose an AttSE branch based on group-aware contrastive learning, which models intra-group relationships by sharing retrieval cues within each semantic group.
- We construct two new OV-PAR benchmark datasets and comprehensively evaluate 20 SOTA methods, demonstrating that our framework achieves the best overall performance on both datasets.

II. Related Work

A. Pedestrian Attribute Recognition

Existing PAR methods fall into three main categories: vision-only [25]–[27], attribute relation modeling [28], [29], and multimodal approaches [30]. To handle large spatial variations among attributes, Shen et al. [31] design an adaptive feature selection strategy using attribute-specific priors for location and scale estimation, while Wu et al. [32] suppress high-frequency activations to enhance discriminative focus. To exploit inter-dependent attributes, Jia et al. [33] build a group-shared attention memory bank for long-tail attributes, and Jin et al. [23] employ part-based queries to aggregate intra-group features. Recent multimodal methods jointly model attribute-attribute and attribute-visual dependencies. Wang et al. [16] propose a region-aware vision-language co-learning framework, and Wu et al. [15] develop image-conditioned textual prompting for fine-grained vision-language reasoning.

Although these methods perform well on PAR datasets, their fixed classifiers limit their applicability in open scenarios. To

overcome this limitation, we propose SRENet, which jointly models spatial, semantic, and attribute relations, significantly enhancing generalization in OV-PAR.

B. Open-Vocabulary Recognition

Existing open-vocabulary recognition methods can be categorized into three main groups: training-free, text-as-image, and vision-language contrastive training methods. Training-free approaches exploit CLIP's strong generalization by designing prompts or region proposals without additional training. Pratt et al. [34] employ LLMs to generate discriminative prompts, while Lin et al. [35] introduce a three-stage coarse-to-global framework. Despite their efficiency, these methods heavily depend on CLIP's feature space and struggle with rare or abstract concepts (e.g., reflective clothing). To address this, text-as-image methods replace visual features with textual representations to reduce training cost. Guo et al. [36] use natural sentences as visual surrogates and learn dual-granularity textual prompts, and Liu et al. [37] leverage LLM-generated text data to train cross-modal classifiers. However, discarding visual cues limits their ability to capture fine-grained details (e.g., shoe types). Recent vision-language contrastive training methods jointly fine-tune both modalities to enhance local semantics. He et al. [20] transfer CLIP's matching ability through distillation, while Tan et al. [21] uses knowledge-constrained optimal transport (OT) for local patch-label matching. SRENet uses OT as one spatial evidence factor and introduces a different transfer mechanism: attribute-conditioned cross-layer selection and base-attribuete labels supervised group learning share the same multi-level aggregation parameters. The resulting group representations are then reused to retrieve configurations containing novel attributes. This learned cross-level coupling distinguishes SRENet from independent local matching or post-hoc propagation of attribute scores.

Recent studies further improve fine-grained vision-language recognition from different perspectives. Wang et al. [38] enrich and reconstruct class names to handle multi-label categories at different semantic granularities. Cai et al. [39] use attention-guided view selection to balance local details and global context in a training-free setting. FG-CLIP [40] improves fine-grained alignment through large-scale region-text supervision and hard negatives. These methods focus on language enrichment, test-time view selection, or foundation-model pretraining, respectively. In contrast, SRENet retains the pretrained CLIP backbone and explicitly decomposes pedestrian-attribute evidence along spatial, cross-layer semantic, and intra-group attribute structures. Related structured-reasoning work studies Bayesian metric objectives [41] or training-free hierarchical scene-graph relations [42].

III. Methodology

In this section, we introduce the proposed SRENet for OV-PAR. We first present its unified overview (Sec. III-A), followed by the spatial-structural enhancement branch (Sec. III-C), semantic-structural enhancement branch (Sec. III-D), and attribute-structural enhancement branch (Sec. III-E). We then describe the shared training objective and score fusion used at inference (Sec. III-F).

A. Overview

SRENet integrates spatial, cross-layer semantic, and intra-group configuration evidence in a shared CLIP space. Given an image I, text representations $T = \{T^{attr}, T^{sen}\}$, and branch-specific structural domains $\mathcal{S}_b$, each enhancement branch predicts scores over the same L_A attributes:

$$\mathcal{P}_b = \Phi_b(I, T; \mathcal{S}_b) \in \mathbb{R}^{L_A}, \qquad b \in \{\text{spa}, \text{sem}, \text{group}\}. \quad (1)$$

SpaSE matches attributes to local patches, while SemSE aggregates attribute-relevant features across encoder layers. Their losses jointly optimize shared visual and attribute text prompts, supporting both fine-grained discrimination and abstract attribute recognition. AttSE uses group prototypes to query the aggregation components shared with SemSE. GCL aligns the resulting group features with base-attribute sentences, supervising this shared path and sentence text prompts. Group-level supervision thus also shapes attribute-level visual aggregation. At inference, AttSE uses the learned group representations to retrieve configurations containing novel attributes. All structural and global branch scores are mapped to a common attribute order and averaged for prediction (Sec. III-F). The following sections provide detailed descriptions of each module.

B. Multimodal Encoder

In this section, our goal is to extract reliable multimodal features to support learning within the dual-branch framework. In OV-PAR, the varying resolutions of pedestrian body parts require precise localization; however, the original CLIP model lacks local alignment supervision and thus struggles to focus on attribute-specific regions. As illustrated in Fig. 2, following CLIP Surgery [43], we adopt a dual-branch visual feature extractor. Detailedly, we insert a consistency-attention CLIP (Con-CLIP) branch after the $\mathbf{K}$-th layer (set to 6). The consistency attention uses value-to-value affinity to build attention map and removes the feed-forward layer. This auxiliary branch operates in parallel with the original CLIP (Ori-CLIP) branch, encouraging each image patch token to attend to semantically similar neighboring regions, thereby enhancing local semantic consistency. Finally, we obtain the original visual features $F^{VO} \in \mathbb{R}^{(N+1)\times D}$ and the consistency-aware visual features $F^{VV} \in \mathbb{R}^{(N+1)\times D}$ from the two branches, respectively. For the text feature extraction branch, we incorporate learnable context prompts [44] to enable dynamic template learning. We introduce two groups of context prompts to learn adaptive textual templates. The resulting text features T^q are extracted using the CLIP text encoder equipped with the corresponding context prompts, where $q \in \{\text{sen}, \text{attr}\}$ denotes group-wise sentences or attributes. We reserve the branch index b for the training losses below to avoid symbol reuse.

C. Spatial-Structural Enhancement

Open-vocabulary pedestrian attribute recognition requires fine-grained grounding because many intra-group attributes

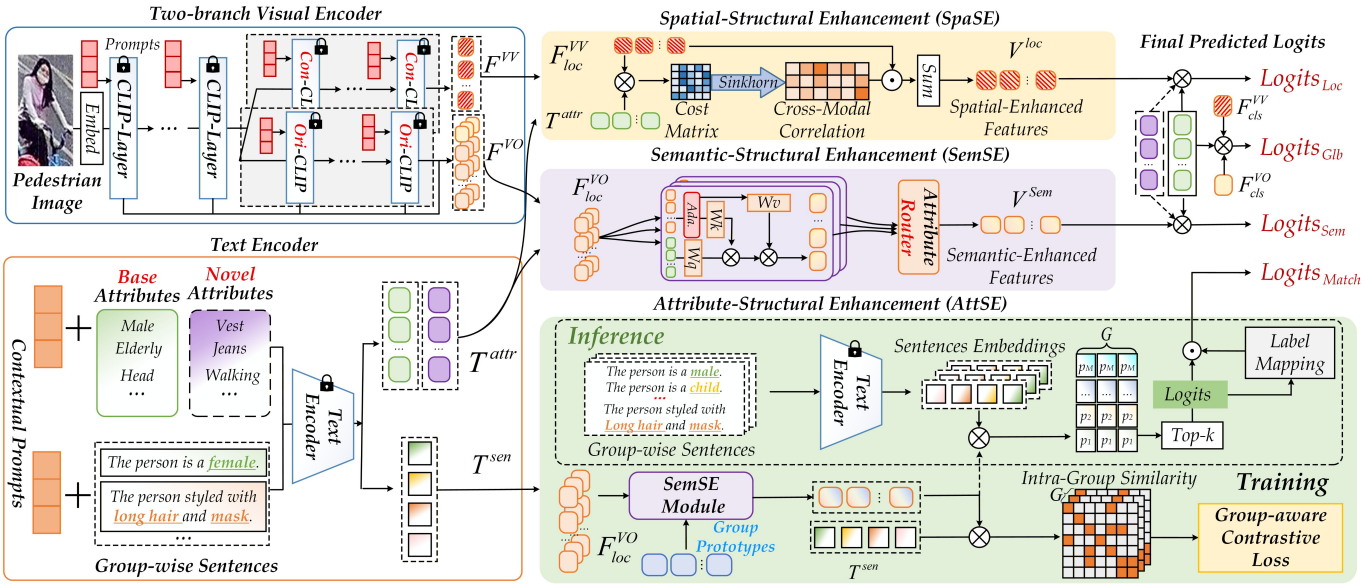


Fig. 2. Overview of SRENet. SpaSE and SemSE produce parallel attribute-level scores from patch-level and cross-layer evidence, respectively. AttSE uses group prototypes to query the multi-level adapters and cross-attention blocks shared with SemSE. GCL trains this shared path using base-attribute sentences, while inference retrieves group configurations that may contain novel attributes. The three structural scores and the global CLIP score are mapped to the same attribute vocabulary and averaged for prediction.

correspond to highly overlapping body regions and differ only subtly (e.g., shoe styles or upper-cloth types). Meanwhile, during training we rely only on supervision from base attributes, which may introduce base bias: the model tends to overemphasize visually salient cues that are frequently covered by base attributes, while overlooking localized discriminative evidence that is crucial for recognizing novel attributes. To explicitly enforce attribute-wise region allocation and obtain interpretable spatial correspondence, we adopt an attribute-fair optimal transport [21], [45] (OT) alignment between visual patch tokens and attribute text embeddings.

Specifically, we define the source distribution $\mathbf{u} \in \mathbb{R}^N$ as a vector of importance weights derived from the maximum attention weight that the attribute query assigns to each image patch, while the target distribution $\mathbf{v}$ represents the importance weights of attributes. To ensure fair treatment of all attributes, $\mathbf{v} = [1/L_A, 1/L_A, \ldots, 1/L_A]^\top \in \mathbb{R}^{L_A}$ is set to a uniform distribution. As described in Sec. III-B, to better localize attribute-relevant regions, we define the cost matrix $\mathbf{C}$ using the negative correlation between $F_{loc}^{VV} \in \mathbb{R}^{N \times D}$ and the attribute text features $T^{attr} \in \mathbb{R}^{L_A \times D}$. Then, the class-balanced OT problem is formulated as:

$$\mathbb{T}^* = \underset{\mathbb{T} \in \Pi(u,v)}{\arg\min} \langle \mathbb{T}, \mathbf{C} \rangle = \underset{\mathbb{T}}{\arg\min} \sum_{i=1}^{N} \sum_{j=1}^{L_A} \mathbb{T}_{ij} \mathbf{C}_{ij}, \tag{2}$$

where $\Pi(u,v) = \{\mathbb{T} \mid \mathbb{T}\mathbf{1}_{L_A} = \mathbf{u}, \mathbb{T}^\top \mathbf{1}_N = \mathbf{v}, \mathbb{T} \geq \mathbf{0}\}$ denotes the set of transport matrices that satisfy the marginal constraints. The OT objective is solved efficiently via the Sinkhorn algorithm [46] with entropy regularization. Finally, the spatial-enhanced features V^{spa} can be formulated as:

$$V^{spa} = \mathrm{sum}(\mathbb{T}^{*\top} \cdot F_{loc}^{VV}) \in \mathbb{R}^{L_A \times D}. \tag{3}$$

Through this process, each attribute automatically obtains equal-quality alignment weights for the most semantically relevant patches, thereby alleviating the model's tendency to overfit base categories while neglecting novel ones.

D. Semantic-Structural Enhancement

SpaSE establishes fine-grained attribute-region correspondences at the local patch level and is therefore particularly effective for low-level appearance attributes such as color, texture, and localized accessories. However, OV-PAR exhibits substantial variation in attribute granularity. Attributes such as gender, age, and action depend more on global context or compositional cues spanning multiple body parts, which are better captured by deeper CLIP representations. Relying only on local-correlation enhancement can therefore be insufficient for these higher-level attributes. SemSE uses each attribute text feature as a query to aggregate related information from the multi-level CLIP features F_{loc}^{VO}. Its routing function $\mathcal{R}(\cdot)$ predicts attribute-specific weights over the selected layers, thereby adapting the balance between local detail and high-level context. The semantically enhanced feature $\mathbf{V}^{sem}$ is formulated as follows:

$$\mathbf{v}_j^{(i)} = \mathcal{CA}(T_j^{attr}, Ada(F_i^{VO})), \quad i \in \mathcal{L}, \tag{4}$$

Here, $Ada(\cdot)$ denotes the adapter used to project visual features into the semantic space, and $\mathcal{CA}(\cdot)$ denotes the cross attention module. Subsequently, we design an attribute routing module $\mathcal{R}(\cdot)$ (a lightweight MLP), that takes the attribute text embedding as input and produces layer-wise fusion weights for each attribute.

$$\alpha_j = \sigma(\mathcal{R}(T_j^{attr})) \in \mathbb{R}^{|\mathcal{L}|}, \quad \sum_{i \in \mathcal{L}} \alpha_{j,i} = 1, \tag{5}$$

Here, σ is the softmax activation function, $\mathcal{L}$ denotes the selected layers of the original CLIP features, which are set

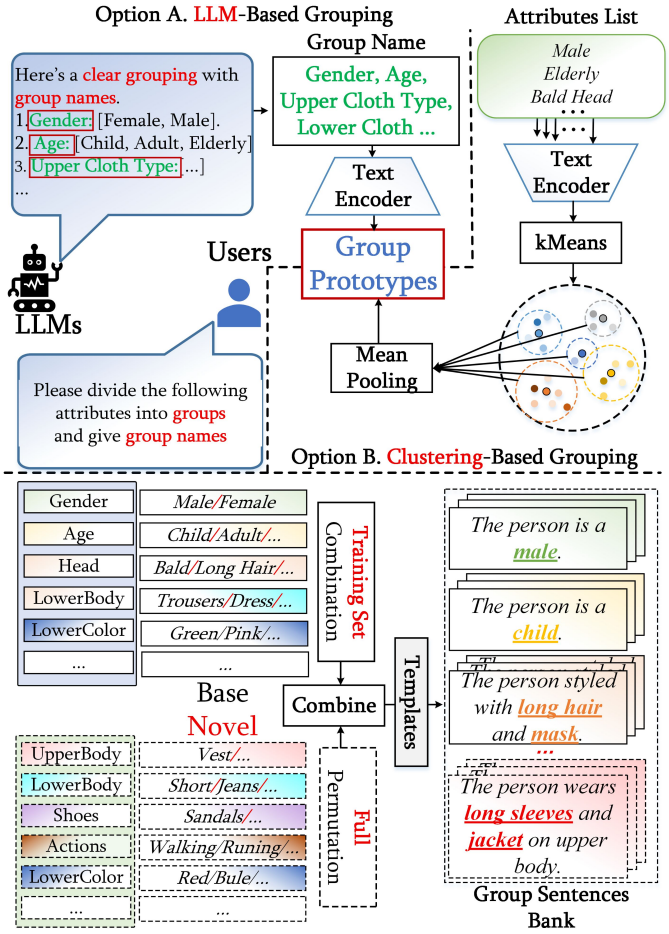


Fig. 3. Offline grouping and sentence-bank construction in AttSE. Option A uses an LLM once to produce semantic groups and group names. Option B removes the LLM and applies k-means to frozen CLIP text embeddings. The resulting grouping is fixed before model training and inference.

to $[2, 5, 8, 11]$ in this work. It then performs a weighted fusion over the cross-layer candidate representations to obtain the final semantically enhanced features.

$$\mathbf{v}_j^{sem} = \sum_{i \in \mathcal{L}} \alpha_{j,i} \cdot \mathbf{v}_j^{(i)}, \quad V^{sem} = [\mathbf{v}_1^{sem}, \ldots, \mathbf{v}_{L_A}^{sem}]^\top. \tag{6}$$

The SemSE does not require manually defined semantic hierarchies or grouping priors for attributes. Instead, the routing weights are driven solely by attribute text semantics and learned end-to-end, enabling hierarchy-aware adaptive semantic enhancement.

E. Attribute-Structural Enhancement

Pedestrian attributes exhibit complex dependency structures: some attributes are mutually exclusive (e.g., *bald* vs. *long hair*), whereas others frequently co-occur (e.g., *formal top* and *formal trousers*); moreover, they often rely on similar localized visual evidence (e.g., *sports shoes* and *casual shoes*). The shared regional evidence provides a basis for transferring knowledge from base to novel attributes within a group. AttSE uses SemSE's aggregation to learn group-conditioned visual representations from base-attribute sentences, then compares these representations with sentence combinations that include novel attributes at inference. Representing intra-group labels jointly as sentences allows retrieval to use their semantic context when predicting individual attributes. Specifically, AttSE constructs the group-level semantic space in three steps:

1) Automatic Attribute Grouping: The AttSE automatically derives an attribute-to-group mapping π via offline automated clustering or LLM-based inference, and then keeps the resulting grouping fixed during both training and inference. Let $\mathcal{A} = \{a_\ell\}_{\ell=1}^{L_A}$ denote the attribute vocabulary. The goal is to obtain a partition mapping:

$$\pi : \mathcal{A} \to \{1, \ldots, G\}, \quad \mathcal{A}^g = \{a \in \mathcal{A} \mid \pi(a) = g\}, \tag{7}$$

where groups are disjoint and cover all attributes. We provide two grouping strategies, which are interchangeable. *Option A: LLM-based semantic grouping.* As shown in Fig. 3, we employ a large language model (LLM) to partition the attribute set into groups and to generate a name for each group. The LLM outputs a grouping $\{\mathcal{A}^g\}_{g=1}^{G}$ along with the corresponding group names. We then encode these names using the text encoder to obtain the group text prototypes p_g. *Option B: CLIP-text-feature clustering.* As a purely model-driven alternative, we cluster attributes in the CLIP text embedding space. For each attribute a_ℓ, we construct a short phrase (e.g., "a photo of a person with {attribute}") and encode it using the frozen CLIP text encoder $\mathrm{E}_T(\cdot)$:

$$t_\ell = \frac{\mathrm{E}_T(a_\ell)}{\|\mathrm{E}_T(a_\ell)\|_2} \in \mathbb{R}^D, \tag{8}$$

We then apply an unsupervised clustering algorithm (e.g., k-means) over $\{t_\ell\}_{\ell=1}^{L_A}$ to obtain $\pi(\cdot)$.

2) Group-Level Sentence Bank Construction: For each group, we construct an offline sentence bank over the group label vector y^g. Its base-attribute component contains only label patterns observed in the training split. For novel attributes, we exhaustively enumerate the Cartesian product of their binary label states within the predefined group, rather than collecting a bank of combinations from evaluation data. The enumeration uses only the attribute vocabulary, fixed grouping, and sentence templates. It does not use test annotations, test frequencies, or test co-occurrence statistics, and it can include configurations absent from the test set. The complete bank is used only for retrieval at inference. During training, each group sentence is instead generated from the observed base-attribute label vector of the current training sample; no sentence containing a novel-attribute state participates in optimization. Each resulting label vector is converted into a natural-language sentence s_i^g and encoded by the frozen CLIP text encoder:

$$\mathcal{E}^g = \{e_i^g\}_{i=1}^{N_g}, e_i^g = \mathrm{E}_T(s_i^g), \tag{9}$$

3) Inference via Group-Level Semantic Retrieval: At inference time, we first derive group-specific visual representations F^g using the group prototype p_g together with SemSE. These representations are matched to candidate sentences containing both base and novel attributes in the shared CLIP space. After L_2 normalization, we compute similarity scores and retrieve the most compatible sentence from each group-level sentence bank:

$$S_i^g = {F^g}^\top e_i^g, \qquad \tilde{S}_i^g = \sigma(S_i^g), \tag{10}$$

We then select the top-K sentences $\mathcal{T}_b^g$ and fuse their label vectors to obtain the group prediction,

$$\hat{y}_b^g = \sum_{i \in \mathcal{T}_b^g} \tilde{S}_{b,i}^g \, y_i^g, \tag{11}$$

Finally, we concatenate predictions from all groups:

$$\mathcal{P}_{group} = \left[\hat{y}^1 \,\|\, \hat{y}^2 \,\|\, \cdots \,\|\, \hat{y}^G \right] \in \mathbb{R}^{L_A}. \tag{12}$$

Notably, the entire inference pipeline is independent of how the groups are obtained (LLM partition or clustering), grouping only determines $\pi(\cdot)$ and the prototypes p^g.

F. Training and Inference

During training, the SpaSE, SemSE, and global branches are supervised by the multi-matching contrastive loss [21] $\mathcal{L}_b$, where $b \in \{\text{spa}, \text{sem}, \text{glb}\}$ identifies the branch output. AttSE uses only base-attribute labels supervision. For every training sample and group, we generate a sentence from that sample's observed base-attribute label vector and encode it as e_i^g. GCL treats other samples in the minibatch with the same base-attribute label vector in group g as positives. Novel-label coordinates are not read, imputed, or used as negatives during training:

$$\mathcal{L}_{gcl} = -\frac{1}{BG} \sum_{i=1}^{B} \sum_{g=1}^{G} \log \frac{\sum_{j \in \mathcal{C}_i^g} \exp(s_{i,j}^g)}{\sum_{k=1}^{B} \exp(s_{i,k}^g)}. \tag{13}$$

Here, $s_{i,j}^g = \cos(F_i^g, e_j^g)/\tau$ denotes the cosine similarity between the group-specific visual feature of sample i and the training sentence generated for sample j. The positive set is $\mathcal{C}_i^g = \{j \mid y_{j,b}^g = y_{i,b}^g\}$, where $y_{i,b}^g$ contains only the base coordinates of group g. We average the image-to-text term above with its symmetric text-to-image counterpart. Because F_i^g is produced by the multi-level adapters and cross-attention blocks shared with SemSE, $\mathcal{L}_{gcl}$ updates the same aggregation path used for attribute-level cross-layer selection rather than fitting a separate AttSE encoder. The three branch classification losses and GCL form the joint objective:

$$\mathcal{L}_{total} = \mathcal{L}_{spa} + \mathcal{L}_{sem} + \mathcal{L}_{glb} + \lambda_g \mathcal{L}_{gcl}. \tag{14}$$

where λ_g denotes the weight of the GCL loss and is set to 1.0 in our method. The final prediction is the parameter-free mean of the four score vectors after they are mapped to the same attribute order:

$$\mathcal{P} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathcal{P}_b, \qquad \mathcal{B} = \{\text{spa}, \text{sem}, \text{glb}, \text{group}\}. \tag{15}$$

SpaSE and SemSE provide parallel attribute-level evidence. The trainable coupling is concentrated between SemSE and AttSE: their shared multi-level adapters and cross-attention blocks receive gradients from both $\mathcal{L}_{sem}$ and $\mathcal{L}_{gcl}$.

IV. Experiments

A. Unified OV-PAR Benchmarks and Evaluation Metric

The cross-dataset benchmark used in POAR [22] combines inconsistent annotation rules, overlooks natural attribute-grouping priors, and includes few mainstream open-vocabulary recognition methods. We construct two unified OV-PAR benchmarks from existing large-scale PAR datasets with consistent attribute definitions, standardized annotation protocols, and explicit base/novel class splits.

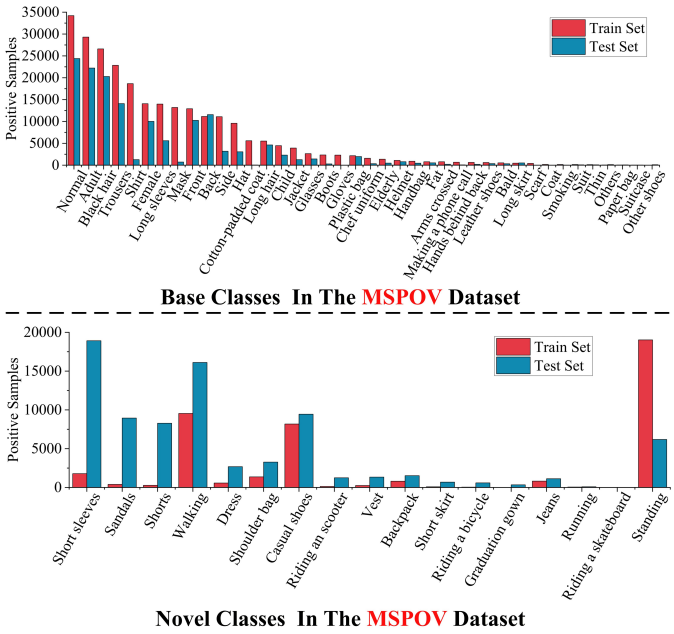


Fig. 4. Base and novel attribute splits in the MSPOV dataset.

MSPOV dataset: MSPOV is constructed based on the cross-domain split of MSP60K [23] for open-vocabulary pedestrian attribute recognition. The dataset contains 35,128 training images and 24,994 testing images, covering 57 attributes in total, of which 40 are treated as base classes and 17 as novel classes.

As shown in Fig. 4, MSPOV inherits the cross-domain training and test partition of MSP60K and measures the positive rate of every attribute in the two domains. The base/novel split is then constructed by jointly considering these train-test distribution differences and the semantic relationships among attributes, rather than by applying a fixed numerical cutoff. Semantically similar or easily confused attributes are assigned to the same subset whenever possible, which avoids separating closely related labels across the base and novel sets. This distribution- and semantics-aware procedure yields 40 base and 17 novel attributes. For example, the related action attributes *Walking*, *Running*, and *Standing* exhibit clear cross-domain distribution differences and are assigned together to the novel set.

RAPOV dataset: RAPOV is reorganized from RAPv2 [24], where the original train/test split is preserved while the attribute definitions and annotation protocols are standardized to support a more systematic evaluation of open-vocabulary pedestrian attribute recognition. The dataset contains 67,943 training images and 16,985 testing images, covering 98 attributes.

These attributes are first grouped by semantics, including identity traits, head appearance, clothing types and colors, shoe type and color, carried items, actions, and occlusion. To obtain 65 base and 33 novel classes, the original RAPOV

TABLE I
COMPARISON ON MSPOV AND RAPOV UNDER ZERO-SHOT LEARNING (ZSL) AND GENERALIZED ZERO-SHOT LEARNING (GZSL), USING MEAN AVERAGE PRECISION (MAP), TOP-3 F1 (OF3), AND TOP-5 F1 (OF5). "-B" USES THE CLIP **VIT-B/16** BACKBONE; † IDENTIFIES RE-IMPLEMENTED METHODS, AND * IDENTIFIES METHODS WHOSE FC CLASSIFIER IS REPLACED WITH A CLIP VIT-B TEXT ENCODER AND LINEAR PROBE.

Method	Pub. Info.	MSPOV						RAPOV					
		ZSL			GZSL			ZSL			GZSL		
		mAP	OF3	OF5	mAP	OF3	OF5	mAP	OF3	OF5	mAP	OF3	OF5
Multimodal Standard PAR Methods													
#1 VTB-B* [47]	TCSVT'22	19.54	36.36	30.04	15.80	44.72	47.03	2.11	4.95	4.10	12.87	24.80	31.19
#2 SSPNet* [31]	PR'24	19.77	41.14	39.08	19.77	37.75	47.78	2.41	5.26	5.07	14.19	14.25	17.39
#3 POAR† [22]	MM'23	19.08	33.96	34.41	15.91	29.78	38.08	2.00	3.44	4.12	12.88	10.53	25.63
#4 PromptPAR-B* [16]	TCSVT'25	20.42	29.38	31.55	26.38	41.84	54.45	4.20	6.17	5.37	42.76	34.62	47.65
#5 EVSITP-B† [15]	CVPR'25	38.30	36.68	34.09	26.43	44.85	55.14	6.08	5.69	5.46	44.55	35.94	51.48
Training-Free Methods													
#1 CLIP [48]	ICML'21	35.86	37.09	40.02	26.19	19.14	25.28	7.51	9.52	8.13	19.00	16.87	22.51
#2 CuPL [34]	ICCV'23	40.63	46.29	48.71	31.06	25.72	32.72	8.94	11.08	9.23	20.45	21.03	28.39
#3 TagCLIP [35]	AAAI'24	42.87	44.97	47.90	30.86	20.74	25.87	9.90	8.25	7.18	24.77	19.90	24.38
#4 CLIP-Surgery [43]	PR'25	41.19	44.25	45.68	28.09	21.50	26.87	10.22	7.50	7.34	24.08	6.27	11.32
#5 SPARC [49]	CVPR'25	41.89	30.59	35.67	31.07	14.66	19.66	10.86	9.90	8.37	23.44	12.06	16.27
Text-as-Image Methods													
#1 TAI-DPT [36]	CVPR'23	32.90	35.00	40.43	24.60	23.35	32.85	10.02	5.44	5.13	26.24	27.47	37.43
#2 CoMC [37]	ICML'24	40.65	22.10	25.80	28.80	11.63	14.33	8.71	8.33	7.82	21.28	13.18	18.64
#3 ProTex [50]	AAAI'25	29.16	31.90	39.24	21.93	12.04	16.92	7.91	7.37	6.84	20.38	10.27	14.72
Vision-Language Fine-Tune Methods													
#1 CoOp [44]	IJCV'22	24.80	38.19	44.33	22.06	38.65	52.26	5.53	5.03	5.96	24.68	32.55	46.63
#2 Co-CoOp [51]	CVPR'22	25.19	41.75	45.28	20.63	34.66	48.72	4.23	7.83	7.48	22.26	29.73	43.21
#3 DualCoOp [52]	NIPS'22	26.89	11.42	15.81	27.66	42.98	54.07	9.84	5.33	5.42	40.37	35.06	49.17
#4 MaPLe [53]	CVPR'23	29.18	31.32	35.44	25.99	40.59	55.09	5.45	7.90	7.99	28.96	33.75	49.42
#5 MKT [20]	AAAI'23	26.96	24.94	34.61	22.74	40.68	49.81	5.79	7.11	6.90	35.97	35.39	49.50
#6 DualCoOp++† [54]	PAMI'24	29.10	17.93	26.26	30.80	43.83	54.97	8.68	6.36	6.35	40.93	35.74	50.42
#7 RAM [21]	CVPR'25	38.83	39.13	38.94	31.75	33.74	48.27	8.99	9.75	7.93	36.50	28.52	42.15
Ours	-	**46.76**	**51.13**	**51.07**	**39.07**	**46.04**	**60.89**	**14.09**	**11.42**	**9.83**	**47.54**	**36.83**	**52.53**

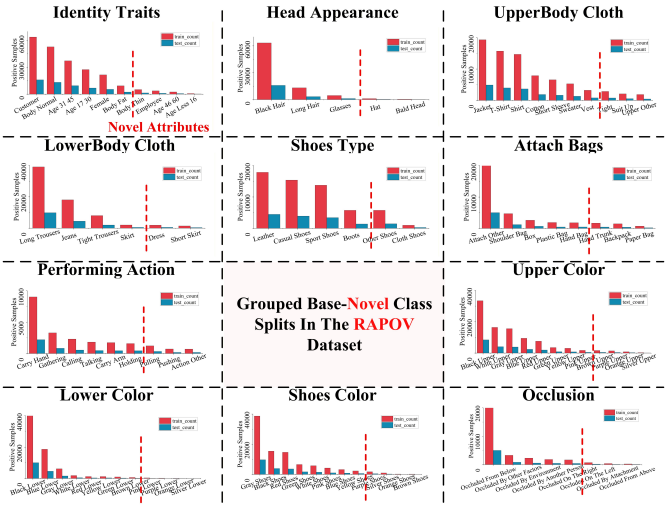


Fig. 5. Grouped base and novel attribute split in the RAPOV dataset.

protocol uses only training-set positive counts: within each group, attributes are sorted by decreasing positive frequency, and approximately the bottom 30% are assigned to the novel set while the remainder form the base set. The part following the red dashed line in Fig. 5 indicate these group-wise tail attributes.

This group-first, then intra-group tail selection strategy ensures that the novel classes are truly sample-sparse and long-tailed, thus forming challenging zero-shot recognition targets. At the same time, maintaining both base and novel classes within each semantic group preserves the semantic structure and interpretability of the dataset. Consequently, RAPOV serves as a suitable benchmark for evaluating multi-level semantic understanding and open-vocabulary generalization in pedestrian attribute recognition.

For both datasets, we conduct an extensive benchmark evaluation of 20 representative methods. These methods span the most influential paradigms in pedestrian attribute recognition and open-vocabulary recognition. For performance assessment, we adopt three widely used metrics: mean Average Precision (mAP) [55], [56], Top-3 F1 score (OF3) [56], [57], and Top-5 F1 score (OF5) [56], [57]. Results are reported under both zero-shot learning (ZSL, evaluated only on novel classes) and generalized zero-shot learning (GZSL, evaluated on both base and novel classes).

B. Implementation Details

We adopt the ViT-B/16 backbone from CLIP [48] and keep both pretrained CLIP encoders frozen. All input images are resized to 224×224, and the maximum number of OT iterations is set to 100. We use VPT-Deep [58] with 10 visual prompts for MSPOV and 15 for RAPOV. The attribute and group-sentence context prompts both have length 32. Training follows two parameter-update stages under the joint objective

in Eq. (14). In the first stage, the text prompts are fixed, while the visual prompts and the multi-level adapters and cross-attention blocks shared by SemSE and AttSE are optimized for 15 epochs using AdamW, with a learning rate of 4×10^{-5} and weight decay of 1×10^{-4}. In the second stage, the first-stage parameters are fixed and only the text prompts are optimized for 20 epochs. All four loss terms in Eq. (14) are evaluated in both stages. The stages differ only in the subset of parameters being updated.

C. Compare with State-of-the-Art Methods

Results on MSPOV Dataset. As shown in Tab. I, our method achieves substantial performance improvements over existing approaches. Compared with CuPL [34], the second-best method with strong generalization ability that generates descriptive captions for given categories using GPT-3 in a training-free manner, our approach achieves significant improvements of +6.13%, +4.84%, and +2.36% on mAP, OF3, and OF5, respectively, under the ZSL setting. The gains are even more substantial under the GZSL setting, reaching +8.01%, +20.32%, and +28.17%, respectively. These consistent and significant gains validate the effectiveness of the SemSE module in aggregating multi-level semantic features and the AttSE branch in modeling attribute correlations, enabling our framework to achieve an excellent balance between generalization to unseen attributes and adaptation to base attributes.

Results on RAPOV Dataset. As shown in Tab. I, compared with previous standard multimodal PAR methods such as PromptPAR [16] and EVSITP [15], our approach achieves a remarkable performance advantage, particularly under the ZSL setting. Specifically, compared with the state-of-the-art EVSITP, our method attains improvements of +8.01%, +5.73%, and +4.37% in mAP, OF3, and OF5, respectively. Furthermore, in comparison with the RAM [21] method, which is also based on OT, our approach achieves gains of +5.10%, +1.65%, and +1.90% on ZSL, and improvements of +11.04%, +8.31%, and +10.38% on GZSL. These results indicate that although both our method and RAM utilize OT to enhance CLIP's generalization capability, our framework achieves a better balance between fitting base attributes and generalizing to novel ones through the integration of the SemSE and AttSE modules.

D. Ablation Study

Table II retains the original main ablation, which reports the incremental construction of SRENet together with the trainable parameter count and memory cost. The progression from the prompt-tuned CLIP baseline to TBV, SpaSE, SemSE, and the complete model shows the overall contribution and computational cost of the proposed design.

We additionally conduct a component-wise ablation study so that each branch is removed while the other two remain unchanged (Tab. III). This complementary protocol directly isolates the marginal contribution of SpaSE, SemSE, and AttSE without discarding the original main ablation or its complexity analysis.

Effect of SpaSE. Removing SpaSE decreases MSPOV ZSL mAP/OF3 by 6.07/8.76 points and GZSL mAP/OF3 by 5.15/2.18 points. On RAPOV, the corresponding decreases are 3.36/5.72 and 4.56/1.29 points. These consistent losses support the role of attribute-fair patch alignment in recovering localized evidence that is not supplied by cross-layer aggregation or group retrieval.

Effect of SemSE. Removing SemSE has a modest effect on MSPOV ZSL (0.13 mAP and 0.89 OF3) but reduces MSPOV GZSL by 2.31 mAP and 7.53 OF3. The reduction is larger on RAPOV GZSL (14.98 mAP and 3.85 OF3), indicating that attribute-conditioned cross-layer aggregation is especially important when base and novel attributes with different semantic granularities are evaluated jointly.

Effect of AttSE. Removing AttSE decreases MSPOV ZSL OF3 by 6.17 points, despite a smaller 0.20-point mAP change, and consistently reduces all RAPOV metrics (1.00/1.35 points under ZSL and 2.14/2.13 under GZSL). This pattern suggests that group-level retrieval primarily improves the consistency of the top-ranked multi-label prediction while still providing complementary gains in average precision.

Efficiency of SRENet. Table IV reports deployment-oriented efficiency measured on a single NVIDIA RTX 4090 in FP32. We time the model forward pass after 50 warm-up iterations, average throughput over 100 repetitions, and exclude data loading and preprocessing. Compared with w/o AttSE, AttSE adds 1.16 ms/image (3.37 vs. 2.21 ms/image), 0.08M trainable parameters, and 0.2G memory, while improving MSPOV ZSL OF3 by 6.17 points and all four RAPOV mAP/OF3 metrics. This is the principal accuracy-efficiency tradeoff of the module. The extra inference cost comes from group-conditioned sentence retrieval, the LLM is used only once offline to produce a fixed grouping and is never invoked during training or inference.

Taken together, the ablations show that the three branches are not interchangeable, SpaSE contributes local spatial evidence, SemSE contributes cross-layer contextual evidence, and AttSE regularizes group-compatible label configurations. Their joint use gives the strongest result on every reported setting.

Impact of Automatic Grouping Strategies. To further investigate the robustness of AttSE with respect to different automatic grouping strategies, we compare LLM-based semantic partitioning with clustering-based alternatives on MSPOV. As shown in Tab. V, both LLM-based grouping and clustering-based grouping consistently outperform the variant without AttSE. When an appropriate number of clusters is adopted (e.g., 12 groups), clustering achieves competitive performance, showing that AttSE can operate in a fully LLM-free setting. In contrast, overly coarse partitions (e.g., 4 groups) degrade performance, suggesting that insufficient semantic granularity harms fine-grained attribute discrimination. This suggests that the grouping strategy is interchangeable, and that the improvement mainly stems from group-conditioned semantic retrieval and group-aware contrastive learning.

Table VI separates the training-time GCL from sentence retrieval at inference. In Simple correlation, we remove GCL and the sentence bank, compute the cosine-similarity matrix between frozen CLIP attribute text embeddings within each

TABLE II
THE ABLATION STUDIES ON THE TWO DATASETS TO EVALUATE THE CONTRIBUTION OF EACH MAJOR MODULE. THE RESULTS REPORT MAP AND OF3 UNDER BOTH ZSL AND GZSL SETTINGS. TBV REFERS TO A DUAL-BRANCH CLIP VISUAL ENCODER EQUIPPED WITH CONSISTENCY ATTENTION BASED ON CLIP-SURGERY [43].

TBV	SpaSE	SemSE	AttSE	MSPOV				RAPOV				Params.	Mem.
				ZSL		GZSL		ZSL		GZSL			
				mAP	OF3	mAP	OF3	mAP	OF3	mAP	OF3		
				36.36	26.90	28.59	35.87	8.18	7.40	28.04	20.82	0.09M	13.7G
✓				38.59	42.04	29.66	42.69	9.31	8.64	32.48	33.18	0.11M	15.0G
✓	✓			43.86	44.81	34.69	41.89	10.96	9.40	41.22	34.71	0.11M	15.1G
✓		✓		41.20	37.36	36.45	45.36	10.89	7.37	43.04	34.85	4.42M	16.5G
✓	✓	✓		46.56	44.96	38.82	44.69	13.09	10.07	45.40	34.70	4.42M	16.5G
✓	✓	✓	✓	**46.76**	**51.13**	**39.07**	**46.04**	**14.09**	**11.42**	**47.54**	**36.83**	4.50M	16.7G

TABLE III
COMPONENT-WISE ABLATION OF SPASE, SEMSE, AND ATTSE ON MSPOV AND RAPOV. RESULTS ARE REPORTED IN MAP AND OF3 UNDER ZSL AND GZSL SETTINGS.

Variant	MSPOV				RAPOV			
	ZSL		GZSL		ZSL		GZSL	
	mAP	OF3	mAP	OF3	mAP	OF3	mAP	OF3
Ours	**46.76**	**51.13**	**39.07**	**46.04**	**14.09**	**11.42**	**47.54**	**36.83**
w/o SpaSE	40.69	42.37	33.92	43.86	10.73	5.70	42.98	35.54
w/o SemSE	46.63	50.24	36.76	38.53	12.78	10.37	32.56	32.98
w/o AttSE	46.56	44.96	38.82	44.69	13.09	10.07	45.40	34.70

TABLE IV
INFERENCE THROUGHPUT ON A SINGLE NVIDIA RTX 4090 USING FP32 AND BATCH SIZE 64. MEASUREMENTS EXCLUDE DATA LOADING AND PREPROCESSING, USE 50 WARM-UP ITERATIONS, AND ARE AVERAGED OVER 100 REPETITIONS.

Method	Images/s	Ms/image
Ours	296.80	3.37
w/o SpaSE	301.18	3.32
w/o SemSE	303.03	3.30
w/o AttSE	451.99	2.21
RAM [21]	363.28	2.75
PromptPAR [16]	376.25	2.66

predefined group, row-normalize this matrix, and use it to linearly propagate the independently predicted attribute scores. Thus, this baseline uses pairwise text correlation but neither generates group sentences nor performs semantic retrieval. Retrieval alone improves ZSL and GZSL OF3 by 2.70 and 1.03 points over the variant without AttSE, while leaving mAP nearly unchanged. It also outperforms Simple correlation on all four metrics, including gains of 1.79 mAP and 1.21 OF3 under GZSL. GCL alone does not improve every metric; its contribution becomes evident when combined with retrieval. Relative to Retrieval only, Full AttSE gains 0.19/3.47 points under ZSL and 0.43/0.32 points under GZSL, indicating that GCL complements retrieval in the complete module.

E. Parameter Analysis

Analysis of the Visual and Text Prompts. As shown in Tab. VII, the model performance improves with an increasing number of prompts and then gradually stabilizes. The best results are achieved when using 5-10 visual prompts, where the improvement in ZSL-OF3 is particularly notable. For textual prompts, performance peaks with 16-32 prompts, while excessive prompts lead to slight fluctuations. Overall, a moderate number of prompts effectively enhances the model's recognition performance under both ZSL and GZSL settings.

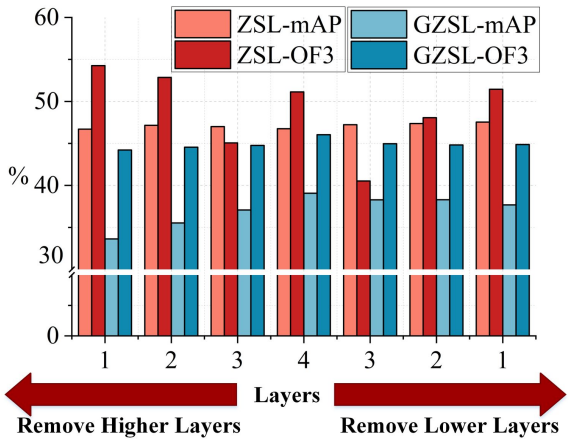


Fig. 6. Effect of CLIP feature layer and insertion position choices in SemSE on MSPOV dataset.

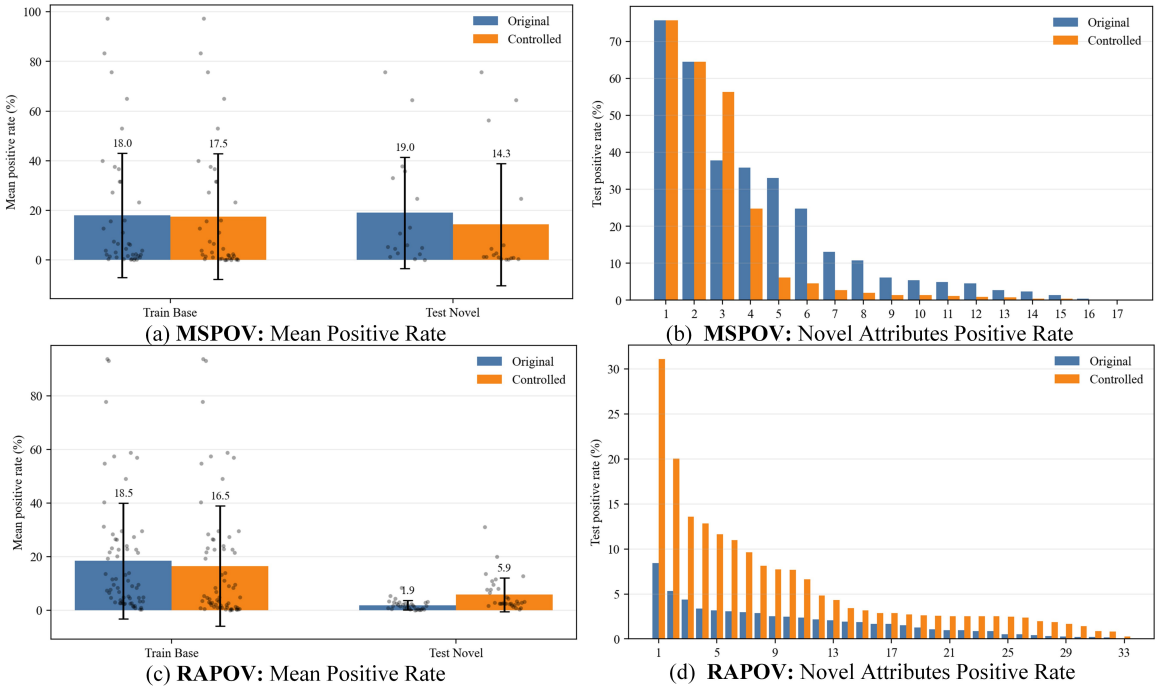
TABLE V
COMPARISON OF DIFFERENT CLUSTERING STRATEGIES ON MSPOV UNDER ZSL AND GZSL SETTINGS. ERROR% DENOTES THE PERCENTAGE OF ATTRIBUTES THAT ARE MISGROUPED RELATIVE TO GROUND-TRUTH.

Cluster Method	Error%	ZSL		GZSL	
		mAP	OF3	mAP	OF3
Ours w/o AttSE	N/A	46.56	44.96	38.82	44.69
Ours w/ AttSE (LLM-based)	0%	**46.76**	**51.13**	**39.07**	**46.04**
AttSE (Cluster 4-Groups)	70.2%	44.37	49.13	35.22	43.92
AttSE (Cluster 8-Groups)	28.1%	45.13	50.01	35.72	44.71
AttSE (Cluster 12-Groups)	17.5%	46.58	50.71	38.65	45.04

TABLE VI
ATTSE MECHANISM ANALYSIS ON MSPOV UNDER ZSL AND GZSL.

Variant	ZSL		GZSL	
	mAP	OF3	mAP	OF3
w/o AttSE	46.56	44.96	38.82	44.69
Simple correlation	46.40	46.59	36.85	44.51
GCL only	46.70	43.31	38.42	44.30
Retrieval only	46.57	47.66	38.64	45.72
Full AttSE	**46.76**	**51.13**	**39.07**	**46.04**

Analysis of the Aggregation Strategies in SpaSE. The results in Tab. VIII demonstrate that different local feature aggre-



(a) **MSPOV:** Mean Positive Rate (b) **MSPOV:** Novel Attributes Positive Rate

(c) **RAPOV:** Mean Positive Rate (d) **RAPOV:** Novel Attributes Positive Rate

Fig. 7. Positive-label distributions before and after controlled attribute selection on MSPOV and RAPOV. In each left panel, bars show the mean positive rate, black markers show individual attribute rates, and error bars show one standard deviation. Each right panel shows the test positive rates of all novel attributes, with the original and controlled sets sorted independently.

TABLE VII
PERFORMANCE COMPARISON FOR PROMPT NUMBERS OF THE VISUAL ENCODER AND THE TEXT ENCODER ON THE MSPOV DATASET.

Prompts	Visual Prompts				Textual Prompts			
	0	5	10	20	0	16	32	64
Zero-Shot Learning								
mAP	45.89	46.84	**46.76**	46.44	47.04	47.33	**46.76**	46.89
OF3	46.10	49.28	**51.13**	48.16	38.02	43.09	**51.13**	46.13
Generalized Zero-Shot Learning								
mAP	36.51	38.46	**39.07**	37.93	37.54	37.79	**39.07**	37.70
OF3	44.10	45.41	**46.04**	45.69	44.28	44.91	**46.04**	45.16

TABLE VIII
COMPARISON OF LOCAL FEATURE AGGREGATION STRATEGIES ON THE MSPOV DATASET.

#	Average	Softmax	Top-k [20]	KCOT [21]	OT
Zero-Shot Learning					
mAP	40.60	44.53	46.10	**46.77**	46.76
OF3	44.94	48.15	48.32	50.56	**51.13**
Generalized Zero-Shot Learning					
mAP	33.93	35.19	36.75	37.98	**39.07**
OF3	45.28	33.44	36.48	45.33	**46.04**

TABLE IX
ABLATION OF THE TARGET ATTRIBUTE MARGINAL IN SPASE ON MSPOV.

Target marginal	ZSL		GZSL	
	mAP	OF3	mAP	OF3
Attribute-fair (uniform)	**46.76**	51.13	**39.07**	**46.04**
Attribute-frequency	45.71	50.41	33.82	44.53
Learnable (uniform init.)	46.57	**51.56**	35.48	45.03
Learnable (frequency init.)	45.75	50.51	33.82	44.48
Unbalanced OT	43.07	35.72	34.53	42.91

TABLE X
COMPARISON ON THE DISTRIBUTION-CONTROLLED MSPOV-C AND RAPOV-C SPLITS.

Dataset	Method	ZSL		GZSL	
		mAP	OF3	mAP	OF3
MSPOV-C	CLIP [48]	32.08	38.15	28.84	21.74
	RAM [21]	29.35	34.20	33.83	35.11
	SRENet	**40.11**	**51.75**	**38.16**	**45.26**
RAPOV-C	CLIP [48]	13.37	17.41	19.50	15.61
	RAM [21]	17.33	12.86	33.74	20.04
	SRENet	**20.79**	**20.74**	**41.62**	**35.12**

gation strategies have a significant impact on performance. Among them, OT and KCOT achieve the best results under both ZSL and GZSL settings, indicating that OT effectively balances the token weight distribution between base and novel categories. However, KCOT [21] relies on teacher guidance during training. While this strategy performs well in natural-scene recognition, it may produce adverse effects in pedestrian scenarios that demand fine-grained understanding, as the original CLIP teacher struggles to distinguish subtle attribute differences.

Analysis of the Attribute Marginal in SpaSE. The uniform target acts as an alignment prior that prevents frequent base attributes from monopolizing the finite patch-to-attribute transport mass within an image. As shown in Tab. IX, it gives the highest ZSL mAP and GZSL mAP/OF3. A learnable marginal initialized uniformly gives a higher ZSL OF3 (51.56 versus 51.13), but its GZSL mAP decreases from 39.07 to 35.48. Frequency-aware and frequency-initialized marginals are less effective, indicating that training frequency can rein-



Fig. 8. Visualization of the attention maps and vision-language correspondence matrices from the SemSE and SpaSE module.

force base-attribute bias. Uniform allocation therefore provides the most stable performance across ZSL and GZSL, whereas the learnable uniform initialization gives the highest ZSL OF3.

Analysis under Controlled Attribute Splits. Because the original MSPOV and RAPOV splits emphasize domain shift and long-tail recognition, they can conflate unseen-attribute transfer with these distributional factors. We therefore construct controlled protocols that preserve the original image partitions and the number of novel attributes in every semantic group. MSPOV-C selects the attributes with the smallest smoothed absolute train-test positive-rate shift within each group, whereas RAPOV-C selects attributes closest to the group-median training frequency instead of group-tail attributes.

Table X shows that SRENet achieves the highest results on both controlled splits. On MSPOV-C, it exceeds the strongest baseline by 8.03/13.60 points in ZSL mAP/OF3 and 4.33/10.15 points in GZSL mAP/OF3. On RAPOV-C, the corresponding margins are 3.46/3.33 and 7.88/15.08 points. Figure 7 visualizes the associated prevalence changes. For the novel attributes selected by each MSPOV protocol, the mean paired absolute train-test positive-rate gap decreases from 14.47 to 10.05 percentage points, a reduction of 4.42 points (30.5%). The average RAPOV test positive rate for novel attributes increases from 1.9% to 5.9%, moving the controlled set away from the extreme frequency tail. For MSPOV, the average training positive rate of base attributes remains close to the original value (18.0% versus 17.5%), while the novel-test distribution changes from 19.0% to 14.3%. SRENet maintains its performance advantage under these controlled attribute distributions.

Analysis of Different Layers in SemSE. As shown in Fig. 6, we analyze the impact of different layer selections in the SemSE module. When the model relies only on low-level or high-level visual features, it shows slight gains in ZSL performance, but results in a significant drop in GZSL. Overall, utilizing all layers provides the best balance between base and novel classes, suggesting that both low-level visual details and high-level semantic cues are crucial for distinguishing attributes with different semantic hierarchies.

F. Visualization

Visualization of ranked matching results. As shown in Fig. 9, we visualize the retrieval results from the AttSE branch

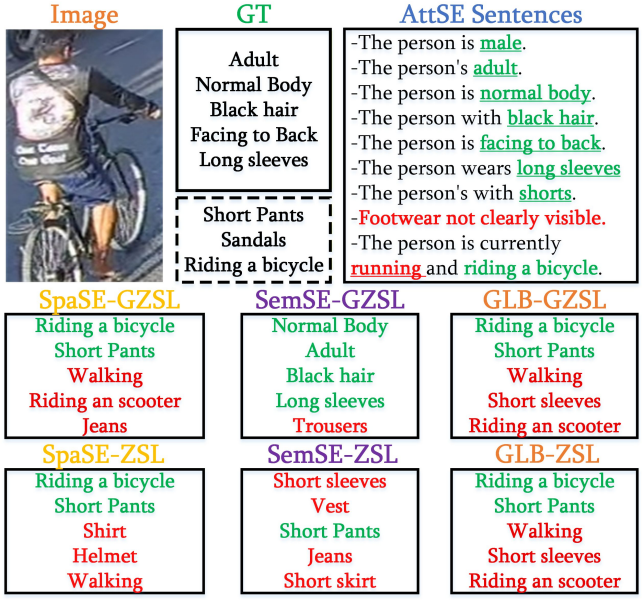


Fig. 9. Visualization of ground-truth and matching results from each branch.

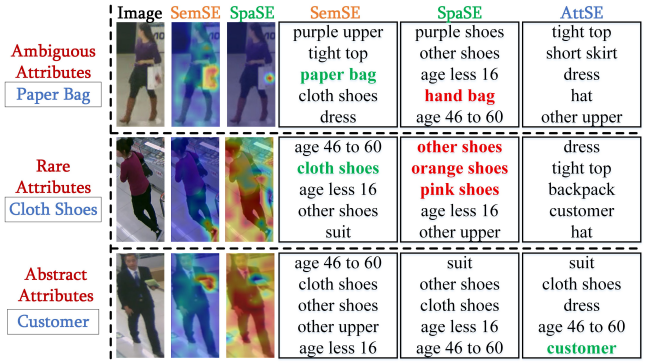


Fig. 10. Prediction results and attention visualization of failure cases.

alongside the matched attributes from the other branches. We observe that the SpaSE branch assigns relatively low scores to global perception attributes, such as body shape and age. In contrast, the SemSE branch accurately identifies these attributes but incorrectly assigns high scores to short sleeves and vests under the ZSL setting. This contrast demonstrates the complementary nature between the two branches. Benefiting from group-shared representations, the AttSE branch successfully retrieves the correct novel-attribute sentences.

Visualization of the attention maps. As shown in Fig. 8, to more clearly demonstrate the distinct roles of the SemSE and SpaSE branches, we visualize their attention maps. We observe that the SemSE branch effectively captures global contextual cues for abstract or high-level attributes (e.g., *Talking*), whereas the SpaSE branch often introduces background noise or fails to localize such attributes accurately. In contrast, the SpaSE branch is able to localize fine-grained local attributes in a more robust manner (such as *Boots*). These results verify that our framework achieves accurate attribute grounding across different semantic levels by jointly enhancing spatial and semantic-structural representations.

V. Conclusion

We presented SRENet for open-vocabulary pedestrian attribute recognition by jointly modeling spatial, semantic, and attribute structures in a shared CLIP space. SpaSE and SemSE learn complementary representations for fine-grained attribute discrimination and abstract attribute recognition. Through the aggregation module shared by SemSE and AttSE, group-level supervision also shapes attribute-level visual aggregation, while group-based sentence retrieval supports knowledge transfer from base to novel attributes. We further established the MSPOV and RAPOV benchmarks for standardized evaluation. Comparisons with 20 representative methods demonstrate the effectiveness of SRENet under both zero-shot and generalized zero-shot settings, and ablation studies confirm the complementary contributions of the three structures.

Recognizing rare attributes and distinguishing visually or semantically similar attributes remain challenging. Future work will explore richer attribute descriptions, sample synthesis to improve rare-attribute coverage, and hierarchical modeling of cross-group dependencies.

References

[1] X. Wang, S. Zheng, R. Yang, A. Zheng, Z. Chen, J. Tang, and B. Luo, "Pedestrian attribute recognition: A survey," *Pattern Recognition*, vol. 121, p. 108220, 2022.

[2] Z. Hu, S. Huang, X. Zhu, F. Sun, B. Zhang, and X. Hu, "Adversarial texture for fooling person detectors in the physical world," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 13 307–13 316.

[3] A. Doering, D. Chen, S. Zhang, B. Schiele, and J. Gall, "Posetrack21: A dataset for person search, multi-object tracking and multi-person pose tracking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 20 963–20 972.

[4] Y. Deng, Z. Chen, C. Li, and J. Tang, "Uncertainty-aware coarse-to-fine alignment for text-image person retrieval," *Visual Intelligence*, vol. 3, no. 1, p. 6, 2025.

[5] S. Yan, N. Dong, L. Zhang, and J. Tang, "Clip-driven fine-grained text-image person re-identification," *IEEE Transactions on Image Processing*, vol. 32, pp. 6032–6046, 2023.

[6] Y. Chen, R. Huang, H. Chang, C. Tan, T. Xue, and B. Ma, "Cross-modal knowledge adaptation for language-based person search," *IEEE Transactions on Image Processing*, vol. 30, pp. 4057–4069, 2021.

[7] Y. Huang, Z. Zhang, Q. Wu, Y. Zhong, and L. Wang, "Attribute-guided pedestrian retrieval: Bridging person re-id with internal attribute variability," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 17 689–17 699.

[8] Y. Zhou, B. Li, H.-M. Hu, X. Zhang, D. Zhang, and H. Wang, "Heterogeneous feature re-sampling for balanced pedestrian attribute recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[9] J. Wu, Y. Huang, M. Gao, Y. Niu, Y. Chen, Q. Wu, and J. Zhao, "Learning comprehensive representation via selective activation and dual-level orthogonality for pedestrian attribute recognition," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 8, pp. 8225–8239, 2025.

[10] X. Fan, Y. Zhang, Y. Lu, and H. Wang, "Parformer: Transformer-based multi-task network for pedestrian attribute recognition," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 1, pp. 411–423, 2023.

[11] Z. Tan, Y. Yang, J. Wan, H. Hang, G. Guo, and S. Z. Li, "Attention-based pedestrian attribute analysis," *IEEE Transactions on Image Processing*, vol. 28, no. 12, pp. 6126–6140, 2019.

[12] X. Liu, H. Zhao, M. Tian, L. Sheng, J. Shao, S. Yi, J. Yan, and X. Wang, "Hydraplus-net: Attentive deep features for pedestrian analysis," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 350–359.

[13] D. Li, Z. Zhang, X. Chen, H. Ling, and K. Huang, "A richly annotated dataset for pedestrian attribute recognition," *arXiv preprint arXiv:1603.07054*, 2016.

[14] Y. Deng, P. Luo, C. C. Loy, and X. Tang, "Pedestrian attribute recognition at far distance," in *Proceedings of the 22nd ACM international conference on Multimedia*, 2014, pp. 789–792.

[15] J. Wu, Y. Huang, M. Gao, Y. Niu, Y. Chen, and Q. Wu, "Enhanced visual-semantic interaction with tailored prompts for pedestrian attribute recognition," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 9570–9579.

[16] X. Wang, J. Jin, C. Li, J. Tang, C. Zhang, and W. Wang, "Pedestrian attribute recognition via clip based prompt vision-language fusion," *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.

[17] Y. Zhou, H.-M. Hu, J. Yu, Z. Xu, W. Lu, and Y. Cao, "A solution to co-occurence bias: attributes disentanglement via mutual information minimization for pedestrian attribute recognition," in *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, ser. IJCAI '23, 2023.

[18] Y. Yang, Z. Tan, P. Tiwari, H. M. Pandey, J. Wan, Z. Lei, G. Guo, and S. Z. Li, "Cascaded split-and-aggregate learning with feature recombination for pedestrian attribute recognition," *International Journal of Computer Vision*, vol. 129, pp. 2731–2744, 2021.

[19] W.-Q. Lu, H.-M. Hu, J. Yu, Y. Zhou, H. Wang, and B. Li, "Orientation-aware pedestrian attribute recognition based on graph convolution network," *IEEE Transactions on Multimedia*, vol. 26, pp. 28–40, 2023.

[20] S. He, T. Guo, T. Dai, R. Qiao, X. Shu, B. Ren, and S.-T. Xia, "Open-vocabulary multi-label classification via multi-modal knowledge transfer," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, no. 1, 2023, pp. 808–816.

[21] H. Tan, Z. Tan, J. Li, A. Liu, J. Wan, and Z. Lei, "Recover and match: Open-vocabulary multi-label recognition through knowledge-constrained optimal transport," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4650–4660.

[22] Y. Zhang, S. Wang, S. Kan, Z. Weng, Y. Cen, and Y.-p. Tan, "Poar: Towards open vocabulary pedestrian attribute recognition," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 655–665.

[23] J. Jin, X. Wang, Q. Zhu, H. Wang, and C. Li, "Pedestrian attribute recognition: A new benchmark dataset and a large language model augmented framework," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 4, 2025, pp. 4138–4146.

[24] D. Li, Z. Zhang, X. Chen, and K. Huang, "A richly annotated pedestrian dataset for person retrieval in real surveillance scenarios," *IEEE Transactions on Image Processing*, vol. 28, no. 4, pp. 1575–1590, 2019.

[25] Y. Zhou, H.-M. Hu, Y. Xiang, X. Zhang, and H. Wu, "Pedestrian attribute recognition as label-balanced multi-label learning," in *Forty-first International Conference on Machine Learning*, 2024.

[26] Z. Tang and J. Huang, "Drformer: Learning dual relations using transformer for pedestrian attribute recognition," *Neurocomputing*, vol. 497, pp. 159–169, 2022.

[27] J. Jia, X. Chen, and K. Huang, "Spatial and semantic consistency regularizations for pedestrian attribute recognition," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 962–971.

[28] Z. Tan, Y. Yang, J. Wan, G. Guo, and S. Z. Li, "Relation-aware pedestrian attribute recognition with graph convolutional networks," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, 2020, pp. 12 055–12 062.

[29] J. Wu, Y. Huang, Z. Gao, Y. Hong, J. Zhao, and X. Du, "Inter-attribute awareness for pedestrian attribute recognition," *Pattern Recognition*, vol. 131, p. 108865, 2022.

[30] J. Jin, X. Wang, Y. Lin, C. Li, L. Huang, A. Zheng, and J. Tang, "Sequencepar: Understanding pedestrian attributes via a sequence generation paradigm," *Pattern Recognition*, p. 112356, 2025.

[31] J. Shen, T. Guo, X. Zuo, H. Fan, and W. Yang, "Sspnet: Scale and spatial priors guided generalizable and interpretable pedestrian attribute recognition," *Pattern Recognition*, vol. 148, p. 110194, 2024.

[32] J. Wu, Y. Huang, M. Gao, Y. Niu, M. Yang, Z. Gao, and J. Zhao, "Selective and orthogonal feature activation for pedestrian attribute recognition," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 6, 2024, pp. 6039–6047.

[33] J. Jia, N. Gao, F. He, X. Chen, and K. Huang, "Learning disentangled attribute representations for robust pedestrian attribute recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, 2022, pp. 1069–1077.

[34] S. Pratt, I. Covert, R. Liu, and A. Farhadi, "What does a platypus look like? generating customized prompts for zero-shot image classification," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 15 691–15 701.

[35] Y. Lin, M. Chen, K. Zhang, H. Li, M. Li, Z. Yang, D. Lv, B. Lin, H. Liu, and D. Cai, "Taglip: A local-to-global framework to enhance open-vocabulary multi-label classification of clip without training," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 4, 2024, pp. 3513–3521.

[36] Z. Guo, B. Dong, Z. Ji, J. Bai, Y. Guo, and W. Zuo, "Texts as images in prompt tuning for multi-label image recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 2808–2817.

[37] Y. Liu, J. Wen, C. Liu, X. Fang, Z. Li, Y. Xu, and Z. Zhang, "Language-driven cross-modal classifier for zero-shot multi-label image recognition," in *Forty-first International Conference on Machine Learning*, 2024.

[38] S. Wang, Q. Wan, J. Gao, and Z. Zeng, "Language-driven multi-label zero-shot learning with semantic granularity," in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 1968–1978.

[39] L. Cai, J. Kang, S. Li, W. Ma, B. Xie, Z. Qin, and J. Liang, "From local details to global context: Advancing vision-language models with attention-based selection," in *Proceedings of the 42nd International Conference on Machine Learning*, vol. 267. PMLR, 13–19 Jul 2025, pp. 6229–6242.

[40] C. Xie, B. Wang, F. Kong, J. Li, D. Liang, G. Zhang, D. Leng, and Y. Yin, "FG-CLIP: Fine-grained visual and textual alignment," in *Forty-second International Conference on Machine Learning*, 2025.

[41] S. Kan, Z. He, Y. Cen, Y. Li, V. Mladenovic, and Z. He, "Contrastive bayesian analysis for deep metric learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 6, pp. 7220–7238, 2022.

[42] W. Liu, L. Cen, Z. Qu, Y. Cen, Y. Liang, Y. Xie, S. Kan *et al.*, "Hsgg: Training-free hierarchical scene graph generation with geometry-guided relation reasoning," in *Forty-third International Conference on Machine Learning*.

[43] Y. Li, H. Wang, Y. Duan, J. Zhang, and X. Li, "A closer look at the explainability of contrastive language-image pre-training," *Pattern Recognition*, vol. 162, p. 111409, 2025.

[44] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *International Journal of Computer Vision*, vol. 130, no. 9, pp. 2337–2348, 2022.

[45] G. Chen, W. Yao, X. Song, X. Li, Y. Rao, and K. Zhang, "PLOT: Prompt learning with optimal transport for vision-language models," in *The Eleventh International Conference on Learning Representations*, 2023.

[46] R. Sinkhorn and P. Knopp, "Concerning nonnegative matrices and doubly stochastic matrices," *Pacific Journal of Mathematics*, vol. 21, no. 2, pp. 343–348, 1967.

[47] X. Cheng, M. Jia, Q. Wang, and J. Zhang, "A simple visual-textual baseline for pedestrian attribute recognition," *IEEE Transactions on Circuits and Systems for Video Technology*, 2022.

[48] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.

[49] K. Miller, A. Gangrade, S. Mishra, K. Saenko, and V. Saligrama, "Sparc: Score prompting and adaptive fusion for zero-shot multi-label recognition in vision-language models," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4313–4321.

[50] M. U. Khattak, M. F. Naeem, M. Naseer, L. Van Gool, and F. Tombari, "Learning to prompt with text only supervision for vision-language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 4, 2025, pp. 4230–4238.

[51] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Conditional prompt learning for vision-language models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16 816–16 825.

[52] X. Sun, P. Hu, and K. Saenko, "Dualcoop: Fast adaptation to multi-label recognition with limited annotations," *Advances in Neural Information Processing Systems*, vol. 35, pp. 30 569–30 582, 2022.

[53] M. U. Khattak, H. Rasheed, M. Maaz, S. Khan, and F. S. Khan, "Maple: Multi-modal prompt learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 19 113–19 122.

[54] P. Hu, X. Sun, S. Sclaroff, and K. Saenko, "Dualcoop++: Fast and effective adaptation to multi-label recognition with limited annotations," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3450–3462, 2024.

[55] Y. Zhang, B. Gong, and M. Shah, "Fast zero-shot image tagging," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2016, pp. 5985–5994.

[56] D. Huynh and E. Elhamifar, "A shared multi-attention framework for multi-label zero-shot learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 8776–8786.

[57] A. Veit, N. Alldrin, G. Chechik, I. Krasin, A. Gupta, and S. Belongie, "Learning from noisy large-scale datasets with minimal supervision," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 839–847.

[58] M. Jia, L. Tang, B.-C. Chen, C. Cardie, S. Belongie, B. Hariharan, and S.-N. Lim, "Visual prompt tuning," in *European conference on computer vision*. Springer, 2022, pp. 709–727.